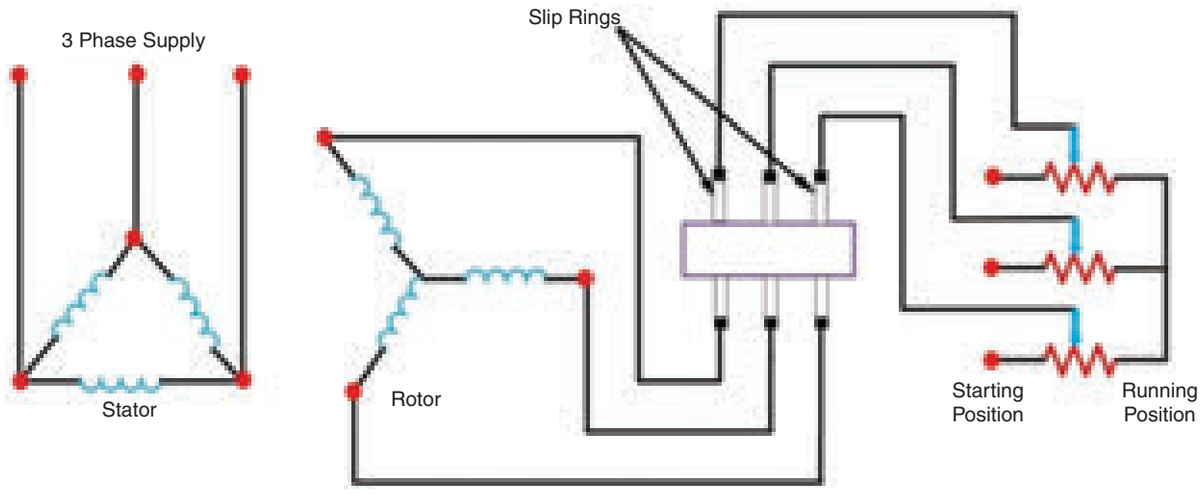


Fig. 8.6 **Rotor resistance starter**

8.4 Electrical isolator

An isolator is a switch intend only for isolating a circuit on no-load. The structure of an electrical isolator is shown in figure 8.7. It should never be used for breaking a circuit carrying appreciable currents. If an isolator is opened carelessly, when carrying a heavy current, the resulting arc could easily cause a flash over to earth. This may shatter the supporting insulators and may even cause a fatal accident to the operator, particularly in high voltage circuits.

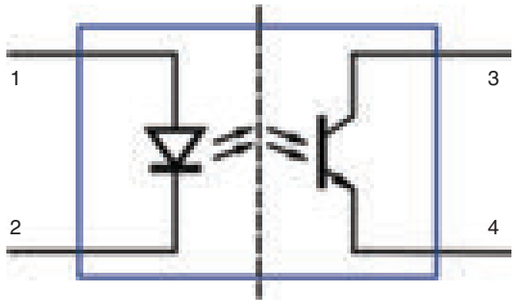


Fig. 8.7 **Electrical isolator**

8.5 Electrical circuit breaker

8.5.1 Description

A circuit breaker is an automatically-operated electrical switch designed to protect an electrical circuit from damage caused by overload of electricity or short circuit. A circuit breaker function is to detect a fault and immediately discontinue electrical flow.

8.5.2 How a circuit breaker works?

A circuit breaker consists of fixed and moving contacts are called electrodes. Under normal operating conditions, these contacts remain closed and do not open automatically until and unless the system becomes faulty. The contacts can be opened manually or by remote control whenever desired. When a fault occurs on any part of the system, the trip coils of the breaker gets energised and the moving contacts are pulled apart by a certain mechanism, thus opening the circuit.

When the contacts of a circuit breaker are separated under faulty conditions, an arc is struck between them. The main problem in a circuit breaker is to extinguish the arc within the shortest possible time so that heat generated by it does not reach a dangerous value.

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